06 Environmental Chemistry

Environmental Pollution:

Environmental pollution is the effect of undesirable changes in our surroundings that cause harmful effects on plants, animals and human beings.

Pollutant:

A substance, which causes pollution, is known as pollutant. Pollutants can be solid, liquid or gaseous substances present in greater concentration than in natural abundance and are produced due to human activities or due to natural happenings.

Atmospheric Pollution:

The lowest region of atmosphere in which the human beings along with other organisms live is called **troposphere.** It extends up to the height of ~ 10 km from sea level.

Above the troposphere, between 10 and 50 km above sea level lies **stratosphere.** The stratosphere contains dinitrogen, dioxygen, ozone and little water vapour.

The presence of ozone in the stratosphere prevents about 99.5% of the Sun's harmful ultraviolet (UV) radiations from reaching the earth's surface

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Tropospheric Pollution :

Tropospheric pollution occurs due to the presence of undesirable solid or gaseous particles in the air.

1. **Gaseous air pollutants:** These are oxides of sulphur, nitrogen and carbon, hydrogen sulphide, hydrocarbons, ozone and other oxidants.
2. **Particulate pollutants:** These are dust, mist, fumes, smoke etc.

1. Gaseous air pollutants :

(a) Oxides of Sulphur : Oxides of sulphur are produced when sulphur containing fossil fuel is burnt.

- Sulphur dioxide is a gas that is poisonous to both animals and plants. It causes respiratory diseases e.g., asthma, bronchitis, emphysema in human beings. Sulphur dioxide causes irritation to the eyes, resulting in tears and redness.
- High concentration of SO_2 leads to stiffness of flower buds which eventually fall off from plants.
- Uncatalysed oxidation of sulphur dioxide is slow. However, the presence of particulate matter in polluted air catalyses the oxidation of sulphur dioxide to sulphur trioxide.

$$2SO_2(g) + O_2(g) \rightarrow 2SO_3(g)$$

The reaction can also be promoted by ozone and hydrogen peroxide.

$$SO_2(g) + O_3(g) \rightarrow SO_3(g) + O_2(g)$$

$$SO_2(g) + H_2O_2(l) \rightarrow H_2SO_4(aq)$$

(b) Oxides of Nitrogen : Dinitrogen and dioxygen do not react with each other at a normal temperature. At high altitudes when lightning strikes, they combine to form oxides of nitrogen.

- NO_2 is oxidised to nitrate ion, $NO_3^{\ominus}$ which is washed into soil, where it serves as a fertilizer.

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- In an automobile engine (at high temperature), when fossil fuel is burnt, dinitrogen and dioxygen combine to yield significant quantities of nitric oxide (NO) and nitrogen dioxide (NO_2) as given below :

$$N_2(g) + O_2(g) \xrightarrow{1483\ K} 2NO(g)$$

NO reacts instantly with oxygen to give NO_2

$$2NO(g) + O_2(g) \rightarrow 2NO_2(g)$$

Rate of production of NO_2 is faster when nitric oxide reacts with ozone in the stratosphere.

$$NO(g) + O_3(g) \rightarrow NO_2(g) + O_2(g)$$

The irritant red haze in the traffic and congested places is due to oxides of nitrogen. Higher concentrations of NO_2 damage the leaves of plants and retard the rate of photosynthesis. It is also a lung irritant.

(c) Hydrocarbons : Hydrocarbons are formed by incomplete combustion of fuel used in automobiles and are carcinogenic, i.e. they cause cancer.

(d) Oxides of Carbon :

(i) Carbon monoxide : It blocks the delivery of oxygen to the organs and tissues. It is produced as a result of incomplete combustion of carbon.

- It binds to haemoglobin to form carboxyhaemoglobin, which is about 300 times more stable than the oxygen-haemoglobin complex. In blood, when the concentration of carboxyhaemoglobin reaches about 3 – 4 per cent, the oxygen carrying capacity of blood is greatly reduced.

(ii) Carbon dioxide : Carbon dioxide (CO_2) is released into the atmosphere by respiration, burning of fossil fuels for energy, and by decomposition of limestone during the manufacture of cement.

- It forms about 0.03 per cent by volume of the atmosphere.
- The increased amount of CO_2 in the air is mainly responsible for global warming.

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Global Warming and Greenhouse Effect:

- About 75 % of the solar energy reaching the earth is absorbed by the earth's surface, which increases its temperature. The rest of the heat radiates back to the atmosphere. Some of the heat is trapped by gases such as carbon dioxide, methane, ozone, chlorofluorocarbon compounds (CFCs) and water vapour in the atmosphere. Thus, they add to the heating of the atmosphere. This causes global warming.
- The atmosphere has kept the temperature on earth constant. As the glass in a greenhouse holds the sun's warmth inside, atmosphere traps the sun's heat near the earth's surface and keeps it warm. This is called natural greenhouse effect.
- Green house gases are carbon dioxide, methane, water vapours, nitrous oxide, CFC's and ozone.

Acid rain :

- Normally rain water has a pH of 5.6 due to the presence of H^+ ions formed by the reaction of rain water with carbon dioxide present in the atmosphere.

$$H_2O(l) + CO_2(g) \rightarrow H_2CO_3(aq)$$

$$H_2CO_3(aq) \rightarrow H^{\oplus}(aq) + HCO_3^{\ominus}(aq)$$

When the pH of the rain water drops below 5.6, it is called acid rain.

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- Burning of fossil fuels (which contain sulphur and nitrogenous matter) such as coal and oil in power stations and furnaces or petrol and diesel in motor engines produce sulphur dioxide and nitrogen oxides. SO_2 and NO_2 after oxidation and reaction with water are major contributors to acid rain.
  $2SO_2(g) + O_2(g) + 2H_2O(l) \rightarrow 2H_2SO_4(aq)$
  $4NO_2(g) + O_2(g) + 2H_2O(l) \rightarrow 4HNO_3(aq)$
- Acid rain is harmful for agriculture, trees and plants as it dissolves and washes away nutrients needed for their growth.
- The Taj Mahal in India has been affected by acid rain.

2. Particulate Pollutants :

- Particulates pollutants are the minute solid particles present in vehicle emissions, smoke particles from fires, dust particles and ash from industries.
- Non-viable particulates may be classified as follows :
  (a) Smoke
  (b) Dust
  (c) Mists
  (d) Fumes
- Lead used to be a major air pollutant emitted by vehicles.

Smog :

- The word smog is derived from smoke and fog. There are two types of smog :
  (a) Classical smog is a mixture of smoke, fog and sulphur dioxide. Chemically it is a reducing mixture and so it is also called as reducing smog.
  (b) Photochemical smog : The main components of the photochemical smog result from the action of sunlight on unsaturated hydrocarbons and nitrogen oxides produced by automobiles and factories. Photochemical smog has high concentration of oxidising agents and is, therefore, called as oxidising smog.

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Formation of photochemical smog :

Two pollutants, hydrocarbons (unburnt fuels) and nitric oxide (NO) when build up to sufficiently high levels, a chain reaction occurs from their interaction with sunlight in which NO is converted into nitrogen dioxide (NO_2). This NO_2 in turn absorbs energy from sunlight and breaks up into nitric oxide and free oxygen atom.

$$NO_2(g) \xrightarrow{1483\ K} NO(g) + O(g) \qquad (i)$$

Oxygen atoms are very reactive and combine with the O_2 in air to produce ozone.

$$O(g) + O_2(g) \rightarrow O_3(g) \qquad (ii)$$

The ozone formed in the above reaction (ii) reacts rapidly with the NO(g) formed in the reaction (i) to regenerate NO_2. NO_2 is a brown gas and at sufficiently high levels can contribute to haze.

$$NO(g) + O_3(g) \rightarrow NO_2(g) + O_2(g) \qquad (iii)$$

Ozone is a toxic gas and both NO_2 and O_3 are strong oxidising agents and can react with the unburnt hydrocarbons in the polluted air to produce chemicals such as formaldehyde, acrolein and peroxyacetyl nitrate (PAN).

$$3CH_4 + 2O_3 \rightarrow \underset{\text{(Formaldehyde)}}{3CH_2O} + 3H_2O$$

$CH_2 = CHCH = O$
Acrolein

CH_3COONO_2 (with O double-bonded to the carbonyl carbon)
Peroxyacetyl nitrate [PAN]

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Both ozone and PAN act as powerful eye irritants. Ozone and nitric oxide irritate the nose and throat and their high concentration causes headache, chest pain, dryness of the throat, cough and difficulty in breathing.

Formation of photochemical smog :

Photochemical smog causes serious health problems.

- Both ozone and PAN act as powerful eye irritants.
- Ozone and nitric oxide irritate the nose and throat and their high concentration causes headache, chest pain, dryness of the throat, cough and difficulty in breathing.
- Photochemical smog leads to cracking of rubber and extensive damage to plant life.
- It also causes corrosion of metals, stones, building materials, rubber and painted surfaces.

Stratospheric Pollution:

Formation and Breakdown of Ozone :

- The upper stratosphere consists of considerable amount of ozone (O_3), which protects us from the harmful ultraviolet (UV) radiations (λ 255 nm) coming from the sun. These radiations cause skin cancer (melanoma) in humans.
- Ozone in the stratosphere is a product of UV radiations acting on dioxygen (O_2) molecules. The UV radiations split apart molecular oxygen into free oxygen (O) atoms. These oxygen atoms combine with the molecular oxygen to form ozone.

$$O_2(g) \xrightarrow{UV} O(g) + O(g)$$

$$O(g) + O_2(g) \underset{}{\overset{UV}{\rightleftharpoons}} O_3(g)$$

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- Ozone is thermodynamically unstable and decomposes to molecular oxygen. Thus, a dynamic equilibrium exists between the production and decomposition of ozone molecules. The main reason of ozone layer depletion the release of chlorofluorocarbon.
- Compounds (CFCs), also known as freons. These compounds are nonreactive, non-flammable, nontoxic organic molecules. Once CFCs are released in the atmosphere, they get broken down by powerful UV radiations, releasing chlorine free radical.

$$CF_2Cl_2(g) \xrightarrow{UV} \dot{Cl}(g) + \dot{C}F_2Cl(g) \qquad (i)$$

The chlorine radical then react with stratospheric ozone to form chlorine monoxide radicals and molecular oxygen.

$$\dot{Cl}(g) + O_3(g) \longrightarrow Cl\dot{O}(g) + O_2(g) \qquad (ii)$$

Reaction of chlorine monoxide radical with atomic oxygen produces more chlorine radicals.

$$Cl\dot{O}(g) + O(g) \longrightarrow \dot{Cl}(g) + O_2(g) \qquad (iii)$$

The chlorine radicals are continuously regenerated and cause the breakdown of ozone.

The Ozone Hole:

- In summer season, nitrogen dioxide and methane react with chlorine monoxide (reaction iv) and chlorine atoms (reaction v) forming chlorine sinks, preventing much ozone depletion, whereas in winter, special type of clouds called polar stratospheric clouds provide surface on which chlorine nitrate formed (reaction iv) gets hydrolysed to form hypochlorous acid (reaction (vi)).
- It also reacts with hydrogen chloride produced as per reaction (v) to give molecular chlorine.

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$$\dot{Cl}O(g) + NO_2(g) \rightarrow ClONO_2(g) \quad \text{(iv)}$$

$$\dot{Cl}(g) + CH_4(g) \longrightarrow \dot{C}H_3(g) + HCl(g) \quad \text{(v)}$$

$$ClONO_2(g) + H_2O(g) \longrightarrow HOCl(g) + HNO_3(g) \quad \text{(vi)}$$

$$ClONO_2(g) + HCl(g) \longrightarrow Cl_2(g) + HNO_3(g) \quad \text{(vii)}$$

When sunlight returns to the Antarctica in the spring, the sun's warmth breaks up the clouds and HOCl and Cl_2 are photolysed by sunlight, as given in reactions (viii) and (ix).

$$HOCl(g) \xrightarrow{h\nu} \dot{O}H(g) + \dot{Cl}(g) \quad \text{(viii)}$$

$$Cl_2(g) \xrightarrow{h\nu} 2\dot{Cl}(g) \quad \text{(ix)}$$

The chlorine radicals thus formed, initiate the chain reaction for ozone depletion.

Water Pollution:

Pollution of water originates from human activities.

Causes of Water Pollution :

(i) Pathogens : The most serious water pollutants are the disease causing agents called pathogens.

(ii) Organic wastes : Organic matter such as leaves, grass, trash etc. pollute water as a consequence of run off. The large population of bacteria decomposes organic matter present in water. They consume oxygen dissolved in water. In cold water, dissolved oxygen (DO) can reach a concentration up to 10 ppm (parts per million).

- The concentration of dissolved oxygen in water is very important for aquatic life. If the concentration of dissolved oxygen of water is below 6 ppm, the growth of fish gets inhibited.
- **Biochemical Oxygen Demand (BOD) :** The amount of oxygen required by bacteria to break down the organic matter present in a certain volume of a sample of water, is called **Biochemical Oxygen Demand (BOD).** Clean water would have BOD value of less than 5 ppm where as highly polluted water could have a BOD value of 17 ppm or more.

(iii) Chemical Pollutants : Water soluble inorganic chemicals that include heavy metals such as cadmium, mercury, nickel etc constitute an important class of pollutants.

Eutrophication :

The process in which nutrient enriched water bodies support a dense plant population, which kills animal life by depriving it of oxygen and results in subsequent loss of biodiversity is known as **Eutrophication.**

International Standards for Drinking Water :

Fluoride : Fluoride ion concentration deficiency in drinking water causes diseases such as tooth decay etc. Soluble fluoride is often added to drinking water to bring its concentration upto 1 ppm or 1 mg dm^{-3}.

- However, $F^{\ominus}$ ion concentration above 2 ppm causes brown mottling of teeth. At the same time, excess fluoride (over 10 ppm) causes harmful effect to bones and teeth. The $F^{\ominus}$ ions make the enamel on teeth much harder by converting hydroxyapatite, $[3(Ca_3(PO_4)_2.Ca(OH)_2]$, the enamel on the surface of the teeth, into much harder fluorapatite, $[3(Ca_3(PO_4)_2.CaF_2]$.

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Lead : The prescribed upper limit concentration of lead in drinking water is about 50 ppb. Lead can damage kidney, liver, reproductive system etc.

Sulphate: Excessive sulphate (>500 ppm) in drinking water causes laxative effect, otherwise at moderate levels it is harmless.

Nitrate: The maximum limit of nitrate in drinking water is 50 ppm. Excess nitrate in drinking water can cause disease such as methemoglobinemia ('blue baby' syndrome).

Table : Maximum Prescribed Concentration of Some Metals in Drinking Water.

Metal	Maximum concentration (ppm or mg dm^{-3})
Fe	0.2
Mn	0.05
Al	0.2
Cu	3.0
Zn	5.0
Cd	0.005

Soil Pollution:

Soil pollution as a part of land degradation is caused by the presence of xenobiotic (human-made) chemicals or other alteration in the natural soil environment.

Pesticides :

- Pesticides are basically synthetic toxic chemicals with ecological repercussions.
- The repeated use of the same or similar pesticides give rise to pests that are resistant to that group of pesticides thus making the pesticides ineffective.

Industrial Waste:

- Industrial solid wastes are also sorted out as biodegradable and non-degradable wastes.
- Biodegradable wastes are generated by cotton mills, food processing units, paper mills, and textile factories.
- Non-biodegradable wastes are generated by thermal power plants which produce fly ash; integrated iron and steel plants which produce blast furnace slag and steel melting slag.
  Nowadays, fly ash and slag from the steel industry are utilised by the cement industry.

Strategies to Control Environmental Pollution:

Waste Management :

- Solid waste is not the only waste, which you see in your household garbage box.
- The improper disposal of wastes is one of the major causes of environmental degradation. Therefore, the management of wastes is of utmost importance.

Swachh Bharat Mission–Urban (SBM–U):

- The SBM–U primarily aims at achieving 100% scientific management of solid waste in the country and making Urban India free from open defecation.

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Swachh Bharat Mission–Gramin (SBM–G) :

- The SBM–G targets to bring about an improvement in the general quality of life in rural areas by promoting cleanliness and hygiene and Eliminating open defecation.

Collection and Disposal :

- Domestic wastes are collected in small bins, which are then transferred to community bins by private or municipal workers.
- Domestic wastes collected in community bins by municipal workers transferred to Disposable Site.
- At these Disposable Sites biodegradable and non-biodegradable waste is separated.
- Biodegradable waste is transferred to get deposited in landfills which naturally get converted into compost.
- Non-biodegradable waste is transferred to get recycled.

Green Chemistry:

Introduction :

- Green chemistry is a production process that would bring about minimum pollution or deterioration to the environment.
- The by-products generated during a process, if not used gainfully, add to the environmental pollution.
- Such processes are not only environmental unfriendly but also cost-ineffective. The waste generation and its disposal both are economically unsound.
- Utilisation of existing knowledge base for reducing the chemical hazards along with the developmental activities is the foundation of green chemistry.

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Green Chemistry in day-to-day Life :

(i) Dry Cleaning of Clothes

Tetra chloroethene ($Cl_2C = CCl_2$) was earlier used as solvent for dry cleaning. The compound contaminates the ground water and is also a suspected carcinogen. Replacement of halogenated solvent by liquid CO_2 will result in less harm to ground water.

(ii) Bleaching of Paper

Chlorine gas was used earlier for bleaching paper. These days, hydrogen peroxide (H_2O_2) with suitable catalyst.

(iii) Synthesis of Chemicals

Ethanal (CH_3CHO) is now commercially prepared by one step oxidation of ethene in the presence of ionic catalyst in aqueous medium with a yield of 90%.

$$CH_2 = CH_2 + O_2 \xrightarrow[Pd(II)/Cu(II)(\text{in water})]{\text{Catalyst}} CH_3CHO\ (90\%)$$

(iv) Green Solution to Clean Turbid Water

Powder of kernel of tamarind seeds has been found to be an effective material to make municipal and industrial waste-water clean. It is non-toxic, biodegradable and cost-effective material. This powder is usually discarded as agricultural waste.

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